Gayatri Vidya Parishad College of Engineering Autonomous

Artificial Intelligence and Machine Learning | Course Code: AI301

Machine Learning Foundations - Machine Learning Mathematical Foundations &

Overview: Ordinary least squares, logistic sigmoid derivation, kernel trick in SVM, cross-entro

I. UNIT-WISE SYLLABUS BREAKDOWN & HIGH-YIELD TOPICS

Unit 1: Statistical Foundations & Linear Models
Maximum Likelihood Estimation MLE, Linear regression, Ridge L2 and Lasso L1 regularization, Batch &

Unit 2: Classification & Probabilistic Models
Logistic regression, Decision boundary derivation, Naive Bayes classifier, Gaussian Discriminant Ana

Unit 3: Support Vector Machines & Kernels
Maximum margin hyperplanes, Lagrange duality, Karush-Kuhn-Tucker KKT conditions, Polynomial and RBF

Unit 4: Ensemble Learning & Clustering
Bootstrap aggregation Bagging, Random Forests, AdaBoost, Gradient Boosted Trees XGBoost, K-Means clu

II. HIGH-YIELD SEMESTER EXAMINATION QUESTIONS

1. Derive the closed-form normal equation theta = X^T X^-1 X^T y for multiple linear regression and state
2. Formulate the dual optimization problem for Support Vector Machines with soft margins and explain slac
3. Describe the mathematical steps of Principal Component Analysis PCA using covariance matrix eigendecom

III. PRESCRIBED TEXTBOOKS & ACADEMIC REPOSITORIES

* Christopher M. Bishop - Pattern Recognition and Machine Learning Springer
* Kevin P. Murphy - Machine Learning: A Probabilistic Perspective MIT Press

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